

Hypothesis Testing with E-Values

Chapter 11: Combining E-Values and Using E-Values as Weights

Based on Ramdas & Wang

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Notation you'll need I

The big picture. This chapter answers three practical questions:

- 1 If I have K separate e-values for the *same* hypothesis, what are *all* the valid ways to combine them into one e-value? (Section 11.2)
- 2 If I have one e-value *and* one p-value for the same hypothesis, how do I combine them? (Section 11.3)
- 3 Can an e-value be used as a “weight” that makes a p-value-based multiple testing procedure (like Benjamini–Hochberg) more powerful? (Section 11.4)

Key vocabulary:

- **E-merging function.** A function $F : \mathbb{R}_+^K \rightarrow \mathbb{R}_+$ that takes K e-values $E_1, \dots, E_K$ (for the same null hypothesis, possibly arbitrarily dependent) and outputs $F(E_1, \dots, E_K)$, which is guaranteed to again be a valid e-value: $\mathbb{E}[F(E_1, \dots, E_K)] \leq 1$ under the null, no matter the dependence structure.
- **Admissible.** An e-merging function F is admissible if no other valid e-merging function G satisfies $G \geq F$ everywhere with $G \neq F$ somewhere. In words: you cannot uniformly improve on it. Admissible rules are the “non-wasteful” ones worth using.

Notation you'll need II

- **Cross-merging.** Combining one e-value E and one p-value P (for the *same* hypothesis) into a single output, which may itself be an e-value or a p-value. Section 11.3 calls such a combining function a *combiner*.
- **E-value as weight.** In multi-hypothesis testing, if hypothesis H_k has both a p-value P_k and an e-value E_k , the quotient P_k/E_k (capped at 1) behaves like a p-value that has been “reweighted” by E_k : large E_k (evidence against H_k) loosens the effective threshold, small E_k tightens it. Section 11.4 plugs these reweighted p-values into ordinary FDR procedures.

Notation you'll need III

Reminder from earlier chapters (Chapter 8). The simplex

$$\Delta_{K+1} = \left\{ \boldsymbol{\lambda} = (\lambda_0, \lambda_1, \dots, \lambda_K) \in \mathbb{R}_+^{K+1} : \sum_{k=0}^K \lambda_k = 1 \right\}$$

indexes a family of linear e-merging functions

$$\mathbb{M}_{\boldsymbol{\lambda}}(e_1, \dots, e_K) := \lambda_0 + \sum_{k=1}^K \lambda_k e_k.$$

Here λ_0 is a constant “skepticism” weight (it shrinks the output toward the trivial e-value 1), and $\lambda_1, \dots, \lambda_K$ are weights on the individual e-values. This chapter’s main goal in Section 11.2 is to prove that *these are the only admissible e-merging functions* (Theorem 8.4, stated already in Chapter 8, proved here).

11.1 Optimal transport: the one-sentence idea I

Plain-language explanation. Optimal transport asks: if you have one pile of sand shaped like distribution μ and want to reshape it into a pile shaped like distribution ν , what is the cheapest way to move the grains, given a cost of moving a grain from x to y ? “Cheapest” is a genuine minimization over all valid ways of moving mass (*couplings*); the dual question – looking at it from the price side – turns out to have an equivalent, often easier-to-compute, formula.

Why it shows up here. Both e-merging functions and p-merging functions must remain valid under *arbitrary dependence* between the input e-values/p-values. Certifying a property “for the worst-case dependence” is exactly a question about the set of all joint distributions (couplings) with fixed marginals – which is precisely multi-marginal optimal transport. We only need the tool, not the theory behind it; we state the duality result without proof.

Setup: the operator $\bigoplus$ and the constraint set D_F

Fix $K \geq 2$. Let $F : [0, \infty)^K \rightarrow \mathbb{R}$ be a bounded Borel function – this will be the “target” function we want to dominate. Let $\mathcal{B}$ be the set of Borel functions on $[0, \infty)$.

Additive-separable upper bounds

For $(\phi_1, \dots, \phi_K) \in \mathcal{B}^K$, define the “sum-of-univariate-functions” operator

$$\left(\bigoplus_{k=1}^K \phi_k \right) (x_1, \dots, x_K) := \sum_{k=1}^K \phi_k(x_k), \quad (x_1, \dots, x_K) \in [0, \infty)^K,$$

and the collection of separable functions that dominate F :

$$D_F = \left\{ (\phi_1, \dots, \phi_K) \in \mathcal{B}^K : \bigoplus_{k=1}^K \phi_k \geq F \right\}.$$

Setup: the operator $\oplus$ and the constraint set D_F II

Intuition. D_F collects every way of writing an upper bound for F as a sum of K functions, each depending on only *one* coordinate. This “separable” structure is exactly what shows up when we bound $\mathbb{E}[F(E_1, \dots, E_K)]$ coordinate by coordinate.

Lemma 11.1: optimal transport duality I

Let $\Gamma(\mu_1, \dots, \mu_K)$ be the set of all Borel measures on $\mathbb{R}^K$ with marginal distributions $\mu_1, \dots, \mu_K$ (all the ways to “glue together” K distributions into a joint one – all possible dependence structures).

Lemma 11.1 (optimal transport duality)

For any bounded Borel $F : [0, \infty)^K \rightarrow \mathbb{R}$ and distributions $\mu_1, \dots, \mu_K$ on $\mathbb{R}$,

$$\sup_{\pi \in \Gamma(\boldsymbol{\mu})} \int F \, d\pi = \inf_{\boldsymbol{\phi} \in D_F} \sum_{k=1}^K \int \phi_k \, d\mu_k, \quad (11.1)$$

where in the infimum we only consider $\boldsymbol{\phi}$ with $\sum_{k=1}^K \int \phi_k \, d\mu_k$ well-defined.

Reading the formula.

- Left side: the worst-case (largest) expectation of F over every possible joint dependence structure π with the given marginals – this is exactly “what’s the worst that arbitrary dependence can do to $\mathbb{E}[F]$?”

Lemma 11.1: optimal transport duality II

- Right side: the cheapest separable upper bound to F , priced out coordinate by coordinate against each marginal.
- Extra regularity: if F is upper semicontinuous, the infimum is attained, and ϕ_k can be chosen upper semicontinuous; if F is decreasing and nonnegative, so can ϕ_k be; if $F \in [0, 1]$, so can ϕ_k .

11.2 Intuition: what are we proving? I

Goal. Recall (from Chapter 8) that an e-merging function F satisfies $\mathbb{E}^{\mathbb{P}}[F(E_1, \dots, E_K)] \leq 1$ for every choice of e-variables $E_1, \dots, E_K$ (each with $\mathbb{E}[E_k] \leq 1$), no matter how they are dependent. We want to show:

Recalling Theorem 8.4

For a function $F : \mathbb{R}_+^K \rightarrow \mathbb{R}_+$:

- (i) If F is an e-merging function, then $F \leq \mathbb{M}_{\lambda}$ for some $\lambda \in \Delta_{K+1}$.
- (ii) F is an *admissible* e-merging function if and only if $F = \mathbb{M}_{\lambda}$ for some $\lambda \in \Delta_{K+1}$.

Why this is a clean and satisfying answer. It says: no matter how clever a merging rule you dream up, it can never beat some simple linear combination $\lambda_0 + \sum_k \lambda_k e_k$. And every point on the simplex Δ_{K+1} gives a genuinely different, genuinely admissible, merging rule. Statement (ii) follows immediately from (i) together with the easy fact that $\mathbb{M}_{\lambda}$ itself is always a valid e-merging function – so all the work is in proving (i).

Warm-up: the case $K = 1$ (Lemma 11.2) I

Intuition first. With a single input, an e-merging function is just a function g of one e-variable that stays a valid e-variable. The claim is that g must be dominated by an *affine* function of x : you cannot do better than mixing a constant with the identity.

Lemma 11.2

Let $\theta \in [1, \infty)$ and $r \in \mathbb{R}_+$. If $g : [0, \theta] \rightarrow \mathbb{R}_+$ satisfies $\mathbb{E}^{\mathbb{P}}[g(E)] \leq r$ for all e-variables E taking values in $[0, \theta]$, then there exists $\lambda \in [0, 1]$ such that

$$g(x) \leq r(1 - \lambda + \lambda x) \quad \text{for } x \in [0, \theta] \cap \mathbb{R}.$$

Proof sketch (the key idea).

- WLOG $r = 1$. First show $g(y) \leq y$ for $y > 1$: if not, a two-point variable putting mass $1/y$ on y (mean 1) would violate $\mathbb{E}[g(E)] \leq 1$. Also $g(x) \leq 1$ for $x \leq 1$.
- Suppose no single λ works. Define $\Lambda_0 = \{\lambda : g(y) > (1 - \lambda) + \lambda y \text{ some } y > 1\}$ and $\Lambda_1 = \{\lambda : g(x) > (1 - \lambda) + \lambda x \text{ some } x < 1\}$. These intervals cover $[0, 1]$, so (by a continuity/sup-inf argument) they must *overlap* at some λ_0 .

Warm-up: the case $K = 1$ (Lemma 11.2) II

- At that overlapping λ_0 , pick violating points $x_0 < 1 < y_0$ and build the two-point e-variable X with $\mathbb{P}(X = y_0) = (1 - x_0)/(y_0 - x_0)$, $\mathbb{P}(X = x_0) = (y_0 - 1)/(y_0 - x_0)$ (mean exactly 1). A direct computation gives $\mathbb{E}[g(X)] > 1$ – a contradiction.

Consequence of Lemma 11.2: no free lunch from transforming E !

Practical takeaway ($r = 1$, general form)

If g is a (dimension-1) e-merging function, there exists $\lambda \in [0, 1]$ with

$$g(x) \leq (1 - \lambda) + \lambda x \quad \text{for all } x \geq 0.$$

Why this matters. Suppose you are handed a single e-variable E and, hoping to do better, you consider transforming it – say $\sqrt{E}$ is also a valid e-variable for any E . Lemma 11.2 says $\sqrt{E}$ is *dominated* by $(1 - \lambda) + \lambda E$ for some λ (here $\lambda = 1/2$ turns out to work). So transforming a single e-value can never uniformly beat a simple affine mix with the constant 1. This is also *why*, in sequential testing (Chapter 7), the only way to combine a new e-variable E_t into a running e-process at each step is via $(1 - \lambda_t) + \lambda_t E_t$ – there is no other admissible option.

Upper semicontinuous reduction (Lemma 11.3) I

Why we need this. The optimal transport duality (Lemma 11.1) gives its nicest guarantees (attained infimum, semicontinuous ϕ_k) when the target function F is upper semicontinuous (u.s.c.). Real e-merging functions need not be u.s.c., so we first replace F by a u.s.c. “majorant” that is at least as good, and show that if we can dominate the majorant, we dominate F too.

Lemma 11.3

If F is an e-merging function, then its upper semicontinuous version

$$F^*(\mathbf{e}) = \lim_{\varepsilon \downarrow 0} F(\mathbf{e} + \varepsilon \mathbf{1}), \quad \mathbf{e} \in [0, \infty)^K,$$

is also an e-merging function, and $F^* \geq F$.

Proof idea. Take e-variables $\mathbf{E}$; for rational $\varepsilon \in (0, 1)$ build a perturbed vector $\mathbf{E}_\varepsilon = (\mathbf{E} + \varepsilon \mathbf{1}) \mathbb{1}_{A_\varepsilon}$ using an independent event A_ε with $\mathbb{P}(A_\varepsilon) = 1 - \varepsilon$. One checks $\mathbb{E}[\mathbf{E}_\varepsilon] \leq \mathbf{1}$, so $F(\mathbf{E}_\varepsilon)$ has expectation ≤ 1 ; expanding gives $\mathbb{E}[F(\mathbf{E} + \varepsilon \mathbf{1})] \leq \frac{1 - \varepsilon F(\mathbf{0})}{1 - \varepsilon}$. Fatou's lemma as $\varepsilon \downarrow 0$ then gives $\mathbb{E}[F^*(\mathbf{E})] \leq 1$.

Proof of Theorem 8.4, step 1: set up the transport problem I

By Lemma 11.3, it suffices to prove the bound for u.s.c. F . Fix $\theta \geq 1$, and let $\mathcal{M}_\varepsilon^\theta$ be the set of all distributions on $[0, \theta]$ with mean ≤ 1 (i.e., distributions of e-variables bounded by θ).

Step 1: rewrite the worst-case expectation as a sup-sup, then apply Lemma 11.1. Since F is bounded, nonnegative on $[0, \theta]^K$, and u.s.c.,

$$\sup_{\mu \in (\mathcal{M}_\varepsilon^\theta)^K} \sup_{\pi \in \Gamma(\mu)} \int F \, d\pi = \sup_{\mu \in (\mathcal{M}_\varepsilon^\theta)^K} \inf_{\phi \in D_F^+} \sum_{k=1}^K \int \phi_k \, d\mu_k, \quad (11.2)$$

where $D_F^+ \subseteq D_F$ restricts to nonnegative, u.s.c. components (available thanks to the regularity clauses of Lemma 11.1).

Interpretation. The left side is: over every joint law π of K e-variables (any dependence, any marginals bounded by θ), what's the worst-case value of $\mathbb{E}[F(\mathbf{E})]$? Since F is an e-merging function, this worst case must be ≤ 1 – that is the constraint we will exploit.

Proof of Theorem 8.4, step 2: swap sup and inf (minimax) I

We now want to swap the order of $\sup_{\mu}$ and $\inf_{\phi}$ in (11.2). Define $J(\mu, \phi) = \sum_{k=1}^K \int \phi_k d\mu_k$, and check three conditions for a minimax theorem (Fan, 1953):

- (i) J is **bilinear** in (μ, ϕ) , hence convex in ϕ and concave in μ .
- (ii) $[0, \theta]$ is compact, so $(\mathcal{M}_{\mathcal{E}}^{\theta})^K$ (weak topology) is tight, hence (Prokhorov) sequentially compact, hence compact.
- (iii) Each ϕ_k u.s.c. $\Rightarrow \mu \mapsto J(\mu, \phi)$ is u.s.c. in μ (weak convergence) for each fixed ϕ .

These conditions license Fan's minimax theorem:

$$\sup_{\mu \in (\mathcal{M}_{\mathcal{E}}^{\theta})^K} \inf_{\phi \in D_F^+} \sum_{k=1}^K \int \phi_k d\mu_k = \inf_{\phi \in D_F^+} \sup_{\mu \in (\mathcal{M}_{\mathcal{E}}^{\theta})^K} \sum_{k=1}^K \int \phi_k d\mu_k. \quad (11.3)$$

Since F is an e-merging function, every term in (11.2) is ≤ 1 ; combining with (11.3),

$$1 \geq \inf_{\phi \in D_F^+} \sum_{k=1}^K \sup_{\mu_k \in \mathcal{M}_{\mathcal{E}}^{\theta}} \int \phi_k d\mu_k. \quad (11.4)$$

The sup and sum have now been decoupled across coordinates k – this is the whole point of the swap.

Proof of Theorem 8.4, step 3: apply Lemma 11.2 coordinatewise I

Write $T_\phi = \sup_{\mu \in \mathcal{M}_\varepsilon^\theta} \int \phi \, d\mu$. Fix $\varepsilon > 0$; by (11.4) choose $(\phi_1, \dots, \phi_K) \in D_F^+$ with $\sum_{k=1}^K T_{\phi_k} \leq 1 + \varepsilon$.

Now recognize T_{ϕ_k} as exactly the setting of Lemma 11.2: T_{ϕ_k} is the smallest r such that $\mathbb{E}[\phi_k(E)] \leq r$ for every e-variable $E \in [0, \theta]$. Lemma 11.2 then hands us, for each k , a constant $h_{\phi_k} \in [0, 1]$ with

$$\phi_k(x) \leq T_{\phi_k} (1 - h_{\phi_k} + h_{\phi_k} x) \quad \text{for all } x \in [0, \theta].$$

Summing over k (using $\phi \in D_F^+$, i.e., $\bigoplus_k \phi_k \geq F$):

$$F(x_1, \dots, x_K) \leq \sum_{k=1}^K T_{\phi_k} (1 - h_{\phi_k} + h_{\phi_k} x_k), \quad x_1, \dots, x_K \in [0, \theta].$$

The right side is exactly $(1 + \varepsilon) \mathbb{M}_{\lambda_{\varepsilon, \theta}}$ for some $\lambda_{\varepsilon, \theta} \in \Delta_{K+1}$ (normalize the $T_{\phi_k}(\dots)$ coefficients, using $\sum T_{\phi_k} \leq 1 + \varepsilon$).

Finish. Δ_{K+1} is compact, so along a subsequence $\lambda_{\varepsilon, \theta} \rightarrow \lambda_0$ as $\varepsilon \downarrow 0$, $\theta \rightarrow \infty$. Continuity of $\lambda \mapsto \mathbb{M}_\lambda$ gives $F \leq \mathbb{M}_{\lambda_0}$ on all of $\mathbb{R}_+^K$. ■

Running numerical example: merging three e-values I

Setup. Suppose three independent studies of the same null hypothesis produce e-values

$$E_1 = 0.6, \quad E_2 = 2.4, \quad E_3 = 1.1.$$

($E_1 < 1$ is mild evidence *for* the null; $E_2 > 1$ is evidence against it.)

Merging rule 1: the simple average ($\lambda_0 = 0$, $\lambda_k = 1/3$)

$$\mathbb{M}_{\text{avg}} = \frac{1}{3}(E_1 + E_2 + E_3) = \frac{0.6 + 2.4 + 1.1}{3} = \frac{4.1}{3} \approx 1.367.$$

Running numerical example: merging three e-values II

Merging rule 2: a “skeptical” weighted average ($\lambda_0 = 0.4$, $\lambda_k = 0.2$ each)

$$\mathbb{M}_{\lambda} = 0.4 + 0.2(E_1 + E_2 + E_3) = 0.4 + 0.2 \times 4.1 = 0.4 + 0.82 = 1.22.$$

Interpretation. Both are valid e-values (Δ_{K+1} guarantees $\mathbb{E}[\mathbb{M}_{\lambda}] \leq 1$ under the null). Adding a positive intercept λ_0 “hedges” the combination toward the neutral value 1: rule 2 gives a smaller, more conservative combined e-value than the plain average. Both merging rules are admissible; which one is preferable depends on how much weight the analyst wants to place on the possibility that all three studies show nothing.

Figure: visualizing the merge I

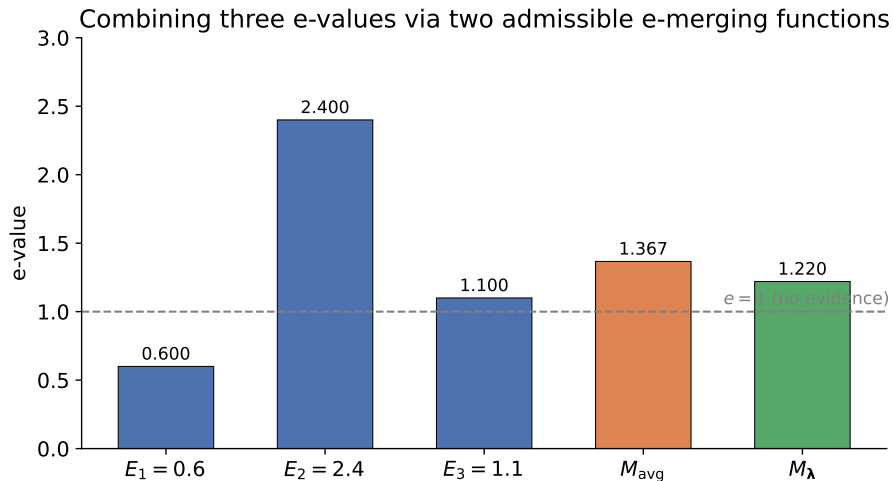


Figure: visualizing the merge II

The three raw e-values (blue) and the two merged e-values: the plain average (orange) and the skeptical, intercept-shrunk average (green). The dashed line at $e = 1$ marks “no evidence.”

Python: computing M_λ e-merging functions

```
import numpy as np

def M_lambda(e, lam0, lam):
    # Admissible linear e-merging function: lam0 + sum_k lam_k * e_k,
    # valid whenever lam0 + sum(lam) == 1 and all weights >= 0 (simplex).
    e, lam = np.asarray(e, float), np.asarray(lam, float)
    assert np.all(lam >= 0) and lam0 >= 0
    assert abs(lam0 + lam.sum() - 1.0) < 1e-9, "weights must lie in the simplex"
    return lam0 + np.dot(lam, e)

E = np.array([0.6, 2.4, 1.1])
M_avg = M_lambda(E, lam0=0.0, lam=np.full(3, 1/3))    # simple average
M_skep = M_lambda(E, lam0=0.4, lam=np.full(3, 0.2))   # skeptical average

print(f"simple average    : {M_avg:.4f}")    # 1.3667
print(f"skeptical average: {M_skep:.4f}")    # 1.2200
```

11.3 Intuition: combining evidence in different currencies I

Sometimes for the *same* hypothesis H , you have one p-value P (say, from a classical test) and one e-value E (say, from a sequential or Bayesian analysis). You want one combined number that is still a valid p-value or e-value. There are four flavors, depending on:

- whether P and E are known/assumed to be **independent** (“i-”) or allowed to be **arbitrarily dependent** (“pe-”), and
- whether the **output** should be a p-value (“/p”) or an e-value (“/e”).

This gives four cases: **i-pe/e**, **i-pe/p**, **pe/e**, **pe/p** (the chapter abbreviates “pe” in these names, e.g. just writing i/e, i/p, /e, /p). A *combiner* is the function that performs this combination.

Definition 11.4: combiners I

Definition 11.4

A function $f : [0, 1] \times [0, \infty] \rightarrow [0, \infty]$ is an **i-pe/e combiner** if $f(P, E)$ is an e-value for any *independent* p-value P and e-value E (for the same null), and $(p, e) \mapsto f(p, e)$ is decreasing in p and increasing in e . Similarly define **i-pe/p**, **pe/p**, **pe/e** combiners, where “i” indicates the independence requirement, and “p” / “e” denote the output type. If the output is a p-value, the combiner is instead increasing in p and decreasing in e .

Monotonicity intuition. A combiner producing an e-value should reward small p (strong evidence) and large e (strong evidence) with a *larger* output; a combiner producing a p-value should do the opposite, since small p-values are the ones signaling evidence.

Four natural combiners I

Let h denote an admissible calibrator (a monotone function turning a p-value into an e-value, from Chapter 2).

The four combiners

- [ie] $C_h^{\text{ie}}(p, e) = h(p) e$ (returns $h(P)E$; convention $0 \times \infty = \infty$). $i\text{-}pe/e$
- [ip] $C^{\text{ip}}(p, e) = (p/e) \wedge 1$ (returns P/E , capped at 1). $i\text{-}pe/p$
- [e] $C_{\lambda, h}^e(p, e) = \lambda h(p) + (1 - \lambda) e$, $\lambda \in (0, 1)$ (average of two e-values $h(P)$ and E). pe/e
- [p] $C^{\text{p}}(p, e) = (2 \min(p, e^{-1})) \wedge 1$. pe/p

Note the pattern. Combiners that assume **independence** ([ie], [ip]) can multiply/divide – multiplication only controls the mean correctly under independence. Combiners valid under **arbitrary dependence** ([e], [p]) must instead use techniques (averaging, min with a factor of 2) that work no matter how P and E co-vary. $C_{\lambda, h}^e$ and C_h^{ie} depend on the chosen calibrator h ; C^{ip} and C^{p} do not – mirroring how there are many admissible p-to-e calibrators but only one admissible e-to-p calibrator.

Theorem 11.5 and the proof of the [ip] case I

Theorem 11.5

For an admissible calibrator h and $\lambda \in (0, 1)$: C_h^{ie} is an admissible i-pe/e combiner; C^{ip} is an admissible i-pe/p combiner; $C_{\lambda, h}^{\text{e}}$ is an admissible pe/e combiner; C^{p} is an admissible pe/p combiner.

Proof that $C^{\text{ip}}(P, E) = P/E \wedge 1$ is a valid p-value (independent case), step by step. For $t \in (0, 1)$,

$$\mathbb{P}(C^{\text{ip}}(P, E) \leq t) = \mathbb{P}(P/E \leq t) = \mathbb{P}(P \leq tE) = \mathbb{E}^{\mathbb{P}}[\mathbb{P}(P \leq tE \mid E)] \leq \mathbb{E}^{\mathbb{P}}[tE] \leq t.$$

Reading each step: (1) unwrap the cap (for $t < 1$ it's inactive); (2) rearrange $P/E \leq t \Leftrightarrow P \leq tE$; (3) tower property, conditioning on E and using independence of P from E ; (4) since P is a valid p-value, $\mathbb{P}(P \leq s) \leq s$ for any constant $s = tE$; (5) $\mathbb{E}[E] \leq 1$ since E is an e-value. This chain of inequalities is exactly why we need **independence**: step (3) requires that conditioning on E doesn't change the distribution of P .

Admissibility is proved by contradiction: assuming some combiner $f \leq C^{\text{ip}}$ but strictly smaller somewhere, one constructs an explicit two-point e-variable E (mirroring the Lemma 11.2 technique) and a uniform P that make $f(P, E)$ violate the p-value property – contradicting validity of f .

The quotient combiner and randomization (Example 11.6) I

$C^{\text{ip}}(P, E) = P/E \wedge 1$ is called the **quotient combiner**. It is the workhorse of Section 11.4.

Example 11.6: Randomization

Let E be an e-variable and $U \sim U[0, 1]$ independent of E . Applying the quotient combiner to (U, E) : U/E is a valid p-value, and it satisfies $U/E \leq 1/E$.

Why this is interesting. The map $e \mapsto (1/e) \wedge 1$ is the *unique* admissible calibrator turning an e-value into a p-value (Chapter 2) – yet here U/E is a strictly *smaller* (more significant) valid p-value than $1/E$ whenever $U < 1$! This shows that allowing external randomization U can improve upon the “best” deterministic e-to-p calibrator. The catch: U/E requires generating extra randomness, which is impractical in general – but becomes practical precisely when applied to an existing, already-random p-value P independent of E , giving the quotient combiner P/E .

When to prefer the quotient combiner? (rule of thumb) I

The quotient combiner P/E is a natural tool for **meta-analysis** combining two independent datasets: one summarized as a p-value P , the other as an e-value E (e.g., because it used optional stopping).

Rule of thumb

- If the two datasets are a priori **exchangeable** (similar power expected), prefer a symmetric method like **Fisher's combination** $P_F := 1 - \chi_4(-2 \log(PP'))$ (using a second p-value P' from the same data instead of E), since asymmetric rules like P/E waste information when both datasets are equally informative.
- If one dataset is **substantially more informative** (the “primary” dataset) than the other (the “secondary” dataset), and the p-value is computed on the more informative one, the quotient combiner P/E can outperform Fisher's combination, and is asymptotically as efficient as the (infeasible) likelihood ratio test based on the full data, both as the significance level $\alpha \rightarrow 0$ and as the secondary sample becomes negligible in size.

A stylized two-sample normal-mean example (comparing $N(0, 1)$ vs. $N(\delta, 1)$) confirms this numerically: when the secondary sample is small relative to the primary, P/E nearly matches the optimal

When to prefer the quotient combiner? (rule of thumb) II

likelihood-ratio test and beats Fisher; when the two samples are balanced, Fisher edges out P/E , and both trail the likelihood ratio test slightly.

11.4 Intuition: e-values as unnormalized weights I

Setting. We now have K hypotheses $H_1, \dots, H_K$, each with a p-value P_k and an e-value E_k (e.g., E_k from a preliminary analysis, P_k from an independent follow-up study; missing values default to 1).

Core idea. Apply the quotient combiner *coordinatewise*:

$$P_k^* := (P_k / E_k) \wedge 1.$$

- If $E_k > 1$ (evidence against H_k), then $P_k^* < P_k$ – the reweighted p-value looks *more* significant, i.e., the e-value strengthens the signal.
- If $E_k < 1$ (no evidence against H_k), then $P_k^* > P_k$ – the reweighted p-value looks *less* significant.

So E_k behaves exactly like a classical multiple-testing **weight**, except it need not be fixed in advance or normalized to sum to K – it can be random and data-dependent, and it can come from a completely separate analysis.

Definition 11.9: the e-weighted p-value procedure I

Definition 11.9 ($\text{ep-}\mathcal{D}$)

Let $\text{p-}\mathcal{D}$ be any multiple testing procedure that takes p-values as input (e.g., Benjamini–Hochberg). The **e-weighted p-value procedure** $\text{ep-}\mathcal{D}$ proceeds as follows: for each $k \in \mathcal{K} = \{1, \dots, K\}$, compute the quotient combiner $P_k^* := (P_k/E_k) \wedge 1$, then supply the vector $\mathbf{P}^* = (P_1^*, \dots, P_K^*)$ to $\text{p-}\mathcal{D}$.

The chapter focuses on **ep-BH**: apply Definition 11.9 with $\text{p-}\mathcal{D}$ = the Benjamini–Hochberg (BH) procedure at level α (Chapter 9). Recall BH controls FDR at level α when the p-values are **PRDS** (a positive-dependence condition, Chapter 9's Definition 9.15).

Theorem 11.10: FDR control for ep-BH I

Let $\mathcal{N} \subseteq \mathcal{K}$ index the true null hypotheses, and $K_0 = |\mathcal{N}|$.

Theorem 11.10

Suppose $\mathbf{P}$ is independent of $\mathbf{E}$ and $\mathbf{P}$ satisfies PRDS. Then the ep-BH procedure has $\text{FDR} \leq K_0 \alpha / K$.

Proof, step by step.

- 1 **Condition on $\mathbf{E}$.** Since $\mathbf{E} \perp \mathbf{P}$, conditional on $\mathbf{E} = \mathbf{e}$, ep-BH is exactly a classical **weighted BH procedure** with (now-deterministic) weights $\mathbf{e}$ applied to the still-PRDS p-values $\mathbf{P}$.
- 2 **Apply the known weighted-BH FDR bound** (Ramdas et al., 2019, Thm. 1): for deterministic weights $\mathbf{e}$,

$$\mathbb{E} \left[\frac{F_{\text{ep-}\mathcal{D}}}{R_{\text{ep-}\mathcal{D}}} \mid \mathbf{E} = \mathbf{e} \right] \leq \frac{1}{K} \sum_{k \in \mathcal{N}} e_k \alpha.$$

Theorem 11.10: FDR control for ep-BH II

③ Take expectation over $\mathbf{E}$ (tower property):

$$\text{FDR}(\text{ep-BH}) = \mathbb{E}\left[\frac{F_{\text{ep-}\mathcal{D}}}{R_{\text{ep-}\mathcal{D}}}\right] \leq \mathbb{E}\left[\frac{1}{K} \sum_{k \in \mathcal{N}} E_k \alpha\right] = \frac{\alpha}{K} \sum_{k \in \mathcal{N}} \mathbb{E}[E_k] \leq \frac{\alpha}{K} \sum_{k \in \mathcal{N}} 1 = \frac{K_0 \alpha}{K}.$$

The last inequality uses $\mathbb{E}[E_k] \leq 1$ for every $k \in \mathcal{N}$ (true nulls), since E_k is a valid e-value there. ■

Remark. Unlike classical weighted BH (fixed weights summing to K), the weights E_k here are random, *data-dependent*, and need not be normalized – yet FDR control still holds, with *no* assumption needed on the dependence structure of $\mathbf{E}$ itself.

Concrete example: an e-value loosens a threshold I

Suppose hypothesis H_k has raw p-value P_k and e-value $E_k = 4.0$ (fairly strong independent evidence against H_k from a preliminary study). Consider a nominal rejection threshold $\alpha = 0.05$ (e.g. the relevant BH cutoff at some step).

Reweighted p-value. $P_k^* = (P_k/E_k) \wedge 1 = P_k/4.0$ (as long as $P_k < 4$, no capping). The testing rule “reject if $P_k^* \leq \alpha$ ” becomes:

$$\frac{P_k}{E_k} \leq \alpha \iff P_k \leq \alpha \cdot E_k = 0.05 \times 4.0 = 0.20.$$

Conclusion. The e-value $E_k = 4.0$ *loosens* the effective significance threshold on the raw p-value from 0.05 to 0.20 – a fourfold increase in the region where H_k gets rejected. Symmetrically, an e-value of $E_k = 0.25$ (weak preliminary evidence) would *tighten* the threshold to $0.05 \times 0.25 = 0.0125$.

The next slide visualizes exactly this loosening/tightening effect across a range of weights.

Figure: e-values as weights loosen/tighten thresholds I

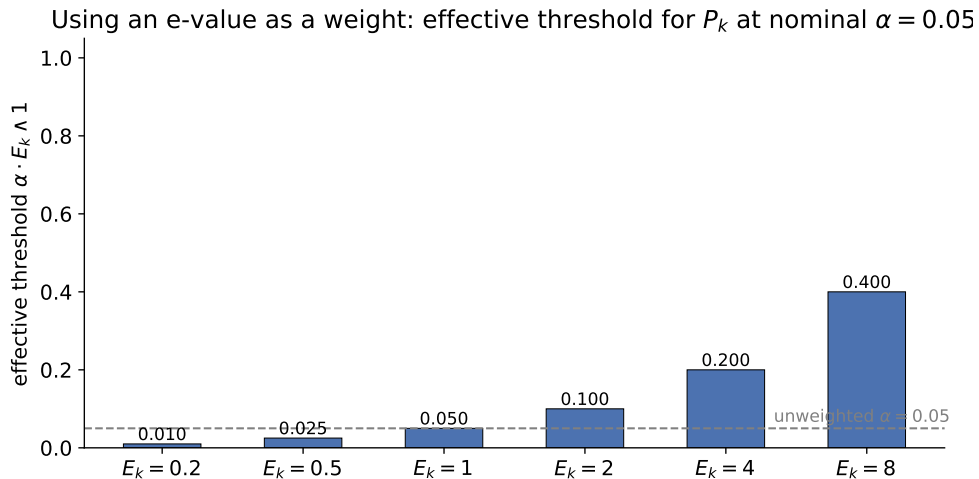


Figure: e-values as weights loosen/tighten thresholds II

Effective rejection threshold $\alpha \cdot E_k \wedge 1$ for a toy panel of six tests with different e-value weights E_k , at nominal $\alpha = 0.05$ (dashed line). Larger E_k loosens the threshold; smaller E_k tightens it.

Python: the ep-BH procedure

```
import numpy as np

def ep_bh(p, e, alpha=0.05):
    # E-weighted BH (ep-BH): Definition 11.9 + Theorem 11.10.
    p, e = np.asarray(p, float), np.asarray(e, float)
    K = len(p)
    p_star = np.minimum(p / e, 1.0) # Step 1: quotient combiner
    order = np.argsort(p_star) # Step 2: ordinary BH on p_star
    sorted_p = p_star[order]
    thresh = alpha * (np.arange(1, K + 1) / K)
    below = np.nonzero(sorted_p <= thresh)[0]
    if len(below) == 0:
        return np.zeros(K, dtype=bool)
    reject = np.zeros(K, dtype=bool)
    reject[order[: below.max() + 1]] = True
    return reject

# E_5 = 4.0 loosens hypothesis 5's effective threshold a lot
p = np.array([0.30, 0.02, 0.45, 0.60, 0.18])
e = np.array([1.00, 1.00, 1.00, 1.00, 4.00])
print("effective thresholds:", np.minimum(p / e, 1.0))
print("ep-BH rejections: ", ep_bh(p, e, alpha=0.05))
```

Chapter summary I

- **11.1** Multi-marginal optimal transport duality (Lemma 11.1) is the technical engine: worst-case expectations under arbitrary dependence (sup over couplings) equal cheapest separable upper bounds (inf over coordinatewise functions).
- **11.2** Using this duality plus a minimax swap and the $K = 1$ Lemma 11.2, we proved Theorem 8.4: every e-merging function is dominated by, and every admissible one equals, a linear combination $\mathbb{M}_{\boldsymbol{\lambda}}(\mathbf{e}) = \lambda_0 + \sum_k \lambda_k e_k$ with $\boldsymbol{\lambda} \in \Delta_{K+1}$.
- **11.3** Four natural combiners cross-merge a p-value and an e-value for the same hypothesis, all provably admissible (Theorem 11.5); the **quotient combiner** $P/E \wedge 1$ is the most broadly useful, and is especially powerful for meta-analysis of an imbalanced pair of datasets.
- **11.4** Plugging $P_k^* = (P_k/E_k) \wedge 1$ into any p-value multiple-testing procedure (e.g. BH) turns e-values into valid, unnormalized, data-dependent **weights**: the ep-BH procedure controls FDR at $K_0\alpha/K$ whenever $\mathbf{P} \perp \mathbf{E}$ and $\mathbf{P}$ is PRDS – with *no* assumption on the dependence of $\mathbf{E}$.

Bibliographical note I

- The optimal transport duality of Lemma 11.1 can be verified via Kellerer (1984); formula (11.1) appears in Rachev and Rüschendorf (2006, Thm. 2.1.1) and Rüschendorf (2013, Thm. 2.3). A standard textbook reference on optimal transport is Villani (2009).
- Vovk and Wang (2021) identified all *symmetric* admissible e-merging functions; the general (non-symmetric) case, Theorem 8.4 as presented here, is due to Wang (2025).
- The content of Sections 11.3–11.4 is based on Ignatiadis et al. (2024b); the full proof of Theorem 11.5 is in their Appendix S1. Using deterministic weights for BH FDR control was studied by Benjamini and Hochberg (1997). The e-weighted BH procedure of Section 11.4 was introduced by Ignatiadis et al. (2024b), along with results for many other weighted procedures.